

Arrays Problems

Problem 1: Class Marks Average

A teacher wants to quickly know the average score of her students after a test.

Problem Statement:

Read N marks into an array and compute sum and average.

Sample Input / Output:

Input: 5, 10 20 30 40 50

Output: Sum=150, Average=30

Test Cases:

TC1: 3 -> 5 5 5 => Sum=15 Avg=5

TC2: 4 -> 0 10 20 30 => Sum=60 Avg=15

Viva Questions:

1. What is an array?
2. How is array memory allocated?
3. Difference between int[] and ArrayList?

Problem 2: Find Maximum Element

A sports coach records running times and wants the fastest time.

Problem Statement:

Store times in array and print minimum value.

Sample Input / Output:

Input: 4, 12 10 14 11

Output: Fastest=10

Test Cases:

TC1: 3 -> 9 7 8 => 7

TC2: 5 -> 1 2 3 4 5 => 1

Viva Questions:

1. How do you initialize an array?
2. Default values of int array?
3. Time complexity to find max/min?

Problem 3: Even Number Counter

A shopkeeper counts even numbered tokens for discount.

Problem Statement:

Count how many elements in the array are even.

Sample Input / Output:

Input: 5, 2 3 4 5 6

Output: Even count=3

Test Cases:

TC1: 4 -> 1 3 5 7 => 0

TC2: 4 -> 2 4 6 8 => 4

Viva Questions:

1. How to check even number?
2. What is modulus operator?
3. Can array store mixed types?

Problem 4: Reverse Display

A digital board must display numbers in reverse order.

Problem Statement:

Print array elements in reverse order.

Sample Input / Output:

Input: 4, 1 2 3 4

Output: 4 3 2 1

Test Cases:

TC1: 3 -> 9 8 7 => 7 8 9

TC2: 1 -> 5 => 5

Viva Questions:

1. What is index of last element?
2. For loop vs while loop?
3. Array bounds exception meaning?

Problem 5: Search an Element

A librarian searches for a book ID inside stored list.

Problem Statement:

Perform linear search to find target element and display its position.

Sample Input / Output:

Input: 5, 10 20 30 40 50, target=30

Output: Found at index 2

Test Cases:

TC1: target not present => Not found

TC2: first element => index 0

Viva Questions:

1. What is linear search?
2. Best case complexity?
3. Can arrays grow dynamically?

Problem 6: Second Largest Score

During a quiz, the organizer wants the second highest score for a consolation prize.

Problem Statement:

Find second largest element in array without sorting.

Sample Input / Output:

Input: 5, 10 20 50 40 30

Output: 40

Test Cases:

TC1: 3 -> 5 4 3 => 4

TC2: 4 -> 8 8 7 6 => 7

Viva Questions:

1. How to track two max values?
2. Edge cases to consider?
3. Why avoid sorting sometimes?

Problem 7: Array Rotation

A conveyor belt rotates packages to the left by one position.

Problem Statement:

Rotate array left by one position.

Sample Input / Output:

Input: 5, 1 2 3 4 5

Output: 2 3 4 5 1

Test Cases:

TC1: 3 -> 9 8 7 => 8 7 9

TC2: 1 -> 6 => 6

Viva Questions:

1. What is shifting?
2. Time complexity?
3. How to rotate by k positions?

Problem 8: Frequency Counter

A survey team counts how many times each rating appears.

Problem Statement:

Count frequency of each element.

Sample Input / Output:

Input: 5, 1 2 2 3 1

Output: 1->2, 2->2, 3->1

Test Cases:

TC1: all same => count=n

TC2: all unique => each=1

Viva Questions:

1. What is nested loop?
2. How to optimize using hashmap?
3. Space complexity meaning?

Problem 9: Merge Two Arrays

Two branches submit sales data which must be combined.

Problem Statement:

Merge two arrays into a single array.

Sample Input / Output:

Input: [1 2 3] and [4 5]

Output: 1 2 3 4 5

Test Cases:

TC1: one empty array

TC2: equal size arrays

Viva Questions:

1. How to create new array size?
2. Copy methods in Java?
3. System.arraycopy purpose?

Problem 10: Palindrome Array Check

A security code is valid if numbers read same forward and backward.

Problem Statement:

Check whether array is palindrome.

Sample Input / Output:

Input: 5, 1 2 3 2 1

Output: Palindrome

Test Cases:

TC1: 1 2 2 1 => yes

TC2: 1 2 3 => no

Viva Questions:

1. What is two-pointer technique?
2. Why compare first and last?
3. Applications of palindrome logic?